

Robocopy Home Lab — Windows CMD

Cybersecurity Focus: Penetration Testing • Cloud Security • Digital Forensics
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1. Lab Description

Overview. This lab covers the practical application of the Windows robocopy utility within CMD, examined from three distinct security perspectives: penetration testing, cloud security, and digital forensics.

Why CMD only? Familiarity with CMD—not just PowerShell—is operationally important. In constrained environments such as secure boot configurations, locked-down enterprise workstations, or recovery consoles, PowerShell may be unavailable or policy-restricted. Robocopy ships natively with every Windows installation and requires no additional tooling.

2. System Information

Field	Value
VM OS Version	Windows 11 Enterprise Evaluation
VM Hypervisor	Oracle VirtualBox v7.2.8 r173730
Cloud Destination	Z:\CloudMigration (C:\CyberLab\CloudMigration)
Lab Date	05 June 2026

3. Environment Preparation

All commands were executed in CMD opened as Administrator. The following subsections document the creation of each file-system component used across all three scenarios.

3.1 Creating the Folder Structure

The lab directory tree was initialised on the C: volume, with Z:\CloudMigration representing the simulated cloud destination via a VirtualBox shared folder. The structure mirrors a minimal on-premises Windows file server.

MKDIR COMMANDS AND DESTINATION VOLUME VERIFICATION

```
C:\>mkdir C:\CyberLab\Staging\Configs
C:\>mkdir C:\CyberLab\Staging\Logs
C:\>mkdir C:\CyberLab\Staging\UserData
C:\>mkdir C:\CyberLab\Staging\Temp
C:\>mkdir C:\CyberLab\ForensicsZone
C:\>dir Z:\
Volume in drive Z is VBOX_VMSHARE
Volume Serial Number is DCA2-37C2

Directory of Z:\

06/04/2026  07:56 PM    <DIR>          .
06/04/2026  07:56 PM    <DIR>          ..
06/04/2026  07:56 PM    <DIR>          CloudMigration
                0 File(s)              0 bytes
                3 Dir(s)  23,389,290,496 bytes free
```

3.2 Server Configuration: db_config.ini

.ini files are general-purpose application configuration files. This file is considered sensitive for the following reasons:

- host=192.168.1.100 — exposes the internal database host IP address
- port=5432 — identifies the service as PostgreSQL, narrowing an attacker's targeting effort
- password — if stored in plaintext, this represents a critical credential exposure

Structure: Section header (e.g. [Database]) followed by key–value pairs (e.g. port=5432).

DB_CONFIG.INI: CREATION AND FILE CONTENT

```
C:\>echo [Database] > "C:\CyberLab\Staging\Configs\db_config.ini"
C:\>echo host=192.168.1.100 >> "C:\CyberLab\Staging\Configs\db_config.ini"
C:\>echo port=5432 >> "C:\CyberLab\Staging\Configs\db_config.ini"
C:\>echo password=REDACTED_IN_VAULT >> "C:\CyberLab\Staging\Configs\db_config.ini"
C:\>type C:\CyberLab\Staging\Configs\db_config.ini
[Database]
host=192.168.1.100
port=5432
password=REDACTED_IN_VAULT
```

3.3 Server Configuration: app.env

.env files store environment variables loaded at application start. This file exposes three high-value artefacts:

- SERVER_URL — reveals internal infrastructure naming conventions and potentially internal-only service endpoints
- API_KEY — the most sensitive field in this file; possession allows an attacker to query APIs, access data, or impersonate the application

- ENV=production — signals that this is a production environment containing real data and business-critical systems

Structure: Key–value pairs only (e.g. API_KEY=sk-internal-abc123).

APP.ENV: CREATION AND FILE CONTENT

```
C:\>echo SERVER_URL=https://internal.corp.local > "C:\CyberLab\Staging\Configs\app.env"
C:\>echo API_KEY=sk-internal-abc123 >> "C:\CyberLab\Staging\Configs\app.env"
C:\>echo ENV=production >> "C:\CyberLab\Staging\Configs\app.env"
C:\>type "C:\CyberLab\Staging\Configs\app.env"
SERVER_URL=https://internal.corp.local
API_KEY=sk-internal-abc123
ENV=production
```

3.4 Authentication Log (auth.log)

The authentication log was populated with a realistic brute-force sequence: one successful authentication followed by five consecutive failures from a single IP address within five seconds, concluding with an account lockout. This pattern is characteristic of automated credential-stuffing or brute-force tooling.

AUTH.LOG: BRUTE-FORCE PATTERN (FIVE FAILED ATTEMPTS FROM 185.220.101.5)

```
C:\>type "C:\CyberLab\Staging\Logs\auth.log"
[2026-06-01 09:12:01] AUTH_SUCCESS user=idoe ip=10.0.0.14
[2026-06-01 10:45:33] AUTH_FAILED user=admin ip=185.220.101.5
[2026-06-01 10:45:34] AUTH_FAILED user=admin ip=185.220.101.5
[2026-06-01 10:45:35] AUTH_FAILED user=admin ip=185.220.101.5
[2026-06-01 10:45:36] AUTH_FAILED user=admin ip=185.220.101.5
[2026-06-01 10:45:37] AUTH_FAILED user=admin ip=185.220.101.5
[2026-06-01 10:46:02] ACCOUNT_LOCKED user=admin
```

3.5 Service Log (system.log)

During incident response and forensic investigations, all available log sources should be reviewed to construct a complete timeline of system activity—addressing when, where, what, and by whom (critical for establishing a defensible chain of custody).

SYSTEM.LOG: NGINX SERVICE START EVENT

```
C:\>type "C:\CyberLab\Staging\Logs\system.log"
[2026-06-01 08:00:01] SVC_START service=nginx pid=1042
```

3.6 Employee File Data (UserData)

These .txt files represent standard employee work documents. Sensitive employee data (PII, SPII) should never reside on a general-purpose work desktop; this directory simulates legitimate business files that require controlled migration.

USERDATA DIRECTORY LISTING

```
C:\>dir C:\CyberLab\Staging\UserData\  
Volume in drive C is Windows  
Volume Serial Number is C48C-B006  
  
Directory of C:\CyberLab\Staging\UserData  
  
06/04/2026  11:05 PM    <DIR>          .  
06/04/2026  09:38 PM    <DIR>          ..  
06/04/2026  11:05 PM                45 Budget_2026.txt  
06/04/2026  11:05 PM                38 HR_Records.txt  
06/04/2026  11:05 PM                38 Q2_Report.txt  
                3 File(s)              121 bytes  
                2 Dir(s)  55,305,678,848 bytes free
```

3.7 Temporary Files (Temp)

.tmp files typically reside in %SYSTEMROOT%\Temp (C:\Windows\Temp) or %TEMP% (C:\Users\<user>\AppData\Local\Temp). This lab focuses on user-scope temporary files. Three representative types are included:

- Lock files — created by applications while a document or resource is in active use (e.g. Microsoft Word)
- Session debuggers — data from an active debugging or application session
- Cached authentication tokens — live credentials stored for session continuity; migration to a new environment provides no value and expands the attack surface

Temporary files should never be migrated: they carry no value in a new environment, and cached tokens in particular can extend the window of compromise if intercepted.

TEMP: LOCK FILE, DEBUG SESSION, AND CACHED AUTHENTICATION TOKEN

```
C:\>echo ~lock_file > "C:\CyberLab\Staging\Temp\~WRD0001.tmp"  
C:\>echo debug_session_id=xF3kP9mR > "C:\CyberLab\Staging\Temp\debug.tmp"  
C:\>echo CACHE_TOKEN=eyJhbGciOiJIUzI1Ni... > "C:\CyberLab\Staging\Temp\session_cache.tmp"
```

4. Scenario 1 — Reconnaissance Dry Run (Penetration Testing)

Context:

Prior to any data exfiltration, a threat actor enumerates the target file system to locate credentials, tokens, and high-value assets. The /L flag instructs robocopy to list all files that would be copied without writing a single byte to disk—this is silent, read-only reconnaissance. The /FP flag appends the full path to each listed item, and /TS includes file timestamps.

Full directory enumeration — mapping the attack surface:

```
robocopy C:\CyberLab\Staging Z:\CloudMigration /E /L /FP /TS
```

OUTPUT: FULL DIRECTORY ENUMERATION WITH TIMESTAMPS AND FULL PATHS

```
ROBOCOPY      ::      Robust File Copy for Windows
-----

Started : Thursday, June 4, 2026 11:53:11 PM
Source   : C:\CyberLab\Staging\
Dest     = Z:\CloudMigration\

Files    : *.*

Options  : *.* /TS /FP /L /S /E /DCOPY:DA /COPY:DAT /R:1000000 /W:30
-----

New Dir      0      C:\CyberLab\Staging\
New File     2      C:\CyberLab\Staging\Configs\
New File      87 2026/06/04 19:19:11 C:\CyberLab\Staging\Configs\app.env
New File      75 2026/06/04 19:01:35 C:\CyberLab\Staging\Configs\db_config.ini
New Dir      2      C:\CyberLab\Staging\Logs\
New File     430 2026/06/04 19:33:03 C:\CyberLab\Staging\Logs\auth.log
New File      57 2026/06/04 19:37:26 C:\CyberLab\Staging\Logs\system.log
New Dir      3      C:\CyberLab\Staging\Temp\
New File      28 2026/06/04 20:18:36 C:\CyberLab\Staging\Temp\debug.tmp
New File      36 2026/06/04 20:18:45 C:\CyberLab\Staging\Temp\session_cache.tmp
New File      13 2026/06/04 20:18:21 C:\CyberLab\Staging\Temp\~WRD0001.tmp
New Dir      3      C:\CyberLab\Staging\UserData\
New File     45 2026/06/04 20:05:54 C:\CyberLab\Staging\UserData\Budget_2026.txt
New File      38 2026/06/04 20:05:43 C:\CyberLab\Staging\UserData\HR_Records.txt
New File      38 2026/06/04 20:05:33 C:\CyberLab\Staging\UserData\Q2_Report.txt
-----

      Total    Copied    Skipped    Mismatch    FAILED    Extras
Dirs  :        5         4         1         0         0         0
Files :       10        10         0         0         0         0
Bytes :       847        847         0         0         0         0
Times :  0:00:00  0:00:00         0:00:00  0:00:00
Ended : Thursday, June 4, 2026 11:53:11 PM
```

The output reveals the complete file-system structure, including file sizes, last-modified timestamps, and full paths—all without initiating a copy operation. The source root directory (C:\CyberLab\Staging) appears as "Skipped" in the summary, which is expected behaviour for the copy source.

Temp folder scan — identifying high-risk files:

```
robocopy C:\CyberLab\Staging\Temp Z:\CloudMigration /E /L
```

OUTPUT: TEMP DIRECTORY SCAN — CACHED TOKEN AND DEBUG SESSION VISIBLE

```
ROBOCOPY      ::      Robust File Copy for Windows
-----

Started : Friday, June 5, 2026 12:05:40 AM
Source   : C:\CyberLab\Staging\Temp\
Dest     = Z:\CloudMigration\

Files    : *.*

Options  : *.* /L /S /E /DCOPY:DA /COPY:DAT /R:1000000 /W:30
-----

New File      3      C:\CyberLab\Staging\Temp\
New File      28      debug.tmp
New File      36      session_cache.tmp
New File      13      ~WRD0001.tmp
-----

      Total    Copied    Skipped    Mismatch    FAILED    Extras
Dirs  :        1         0         1         0         0         0
Files :        3         3         0         0         0         0
Bytes :       77        77         0         0         0         0
Times :  0:00:00  0:00:00         0:00:00  0:00:00
Ended : Friday, June 5, 2026 12:05:40 AM
```

Observations:

- The full directory listing constitutes a complete file-system enumeration—achieved without triggering write-based access alerts

- app.env exposes API_KEY; db_config.ini references a database password—these are primary attacker targets
- session_cache.tmp contains a live authentication token—a high-value lateral movement asset
- Nothing was copied. Nothing was written. The entire file system was mapped.

In a real penetration test, running robocopy /L against a network share (UNC path: \\server\share) performs directory traversal enumeration without triggering most basic file-access alerts, since no copy operations occur.

For defenders: read permissions on sensitive shares must be audited with the same rigour applied to write permissions. Read-only access is sufficient for full enumeration.

5. Scenario 2 — Secure Cloud Migration (Cloud Security)

Context:

The organisation is migrating its on-premises Windows file server to a cloud storage target (Azure Files, AWS FSx, or equivalent). Three requirements apply: (1) exclude the Temp directory, which contains active session tokens; (2) preserve NTFS ACLs so that access controls survive the migration intact; (3) produce an audit log as a compliance record.

5.1 First Migration Attempt — ACL Error on Shared Volume

```
robocopy C:\CyberLab\Staging Z:\CloudMigration /E /XD Temp /SEC
/LOG:C:\CyberLab\migration_audit.log /NP
```

Flag reference: /E — copy all subdirectories including empty ones; /XD Temp — exclude the Temp directory and all contents; /SEC — copy NTFS ACLs from both NTFS and share permissions; /LOG — write output to a log file (overwrites if exists); /NP — suppress percentage progress output.

Note: /XF *.tmp is an alternative to exclude temporary files by extension, but without /XD Temp the directory itself would still be created at the destination.

MIGRATION_AUDIT.LOG

```
C:\>type C:\CyberLab\migration_audit.log

-----
ROBOCOPY      ::      Robust File Copy for Windows
-----

Started : Friday, June 5, 2026 12:31:31 AM
Source  : C:\CyberLab\Staging\
Dest    : Z:\CloudMigration\

Files   : *.*

Exc Dirs : Temp

Options : *.* /S /E /DCOPY:DA /COPY:DATS /NP /R:1000000 /W:30

-----

NOTE : Security may not be copied - Destination might not support persistent ACLs.
```

Well, the problem is there. Now, should review the output till the end.

Result — migration summary:

SUMMARY: 0 FILES COPIED — ONLY DIRECTORIES TRANSFERRED

	Total	Copied	Skipped	Mismatch	FAILED	Extras
<u>Dirs :</u>	<u>5</u>	<u>3</u>	2	0	0	0
<u>Files :</u>	<u>0</u>	<u>0</u>	0	0	0	0
<u>Bytes :</u>	<u>0</u>	<u>0</u>	0	0	0	0
Times :	0:00:00	0:00:00			0:00:00	0:00:00
Ended :	Friday, June 5, 2026 12:31:31 AM					

DESTINATION TREE: FOLDER STRUCTURE PRESENT, NO FILES

```
C:\>tree /f [redacted] VMShare
[redacted]
[redacted]
[redacted]
Full Path [redacted] VMSHARE
├── CloudMigration
│   ├── Configs
│   ├── Logs
│   └── UserData
```

5.2 Root Cause Analysis and Remediation

The root cause was the choice of destination: a VirtualBox VM Shared Folder (Z:\) does not implement the Windows NTFS security model. In shared folder environments, only file data is preserved; ACL metadata is either stripped or translated, causing robocopy to abort file transfers when /SEC is specified and ACL persistence cannot be guaranteed.

Resolution: create a native NTFS destination on the VM's C: volume and re-run the migration.

5.3 Corrected Migration — Native NTFS Volume

```
robocopy C:\CyberLab\Staging C:\CyberLab\CloudMigration /E /XD Temp /SEC
/LOG:C:\CyberLab\migration_audit.log /NP
```

CORRECTED COMMAND: DESTINATION CHANGED TO NATIVE NTFS VOLUME

```
C:\>robocopy C:\CyberLab\Staging C:\CyberLab\CloudMigration /E /XD Temp /SEC
/LOG:C:\CyberLab\migration_audit.log /NP

Log File : C:\CyberLab\migration_audit.log
```

The absence of terminal output is expected—the /LOG flag redirects all status output to migration_audit.log. To display output simultaneously in the terminal and the log file, append the /TEE flag.

Verify migration — audit log review:

MIGRATION_AUDIT.LOG SUMMARY: 7 FILES, 770 BYTES, 2 SKIPPED DIRECTORIES

	Total	Copied	Skipped	Mismatch	FAILED	Extras
<u>Dirs :</u>	<u>5</u>	<u>3</u>	<u>2</u>	0	0	0
<u>Files :</u>	<u>7</u>	<u>7</u>	0	0	0	0
<u>Bytes :</u>	<u>770</u>	<u>770</u>	0	0	0	0
Times :	0:00:00	0:00:00			0:00:00	0:00:00

Summary interpretation: 2 skipped directories — (1) the source root C:\CyberLab\Staging (normal robocopy behaviour) and (2) the excluded Temp directory. Skipped Files: 0 confirms that all non-excluded files were transferred successfully.

Verify Temp directory was excluded:

DESTINATION TREE: CONFIGS, LOGS, AND USERDATA PRESENT — TEMP ABSENT

```
C:\>tree /f C:\CyberLab\CloudMigration
Folder PATH listing for volume Windows
Volume serial number is C48C-B006
C:\CYBERLAB\CLOUDMIGRATION
├── Configs
│   ├── app.env
│   └── db_config.ini
├── Logs
│   ├── auth.log
│   └── system.log
└── UserData
    ├── Budget_2026.txt
    ├── HR_Records.txt
    └── Q2_Report.txt
```

Observations:

- The destination contains Configs\, Logs\, and UserData\ — the Temp directory and all its contents were excluded as required
- session_cache.tmp, debug.tmp, and ~WRD0001.tmp were never written to the destination—preventing token exposure in the target environment
- The audit log provides timestamped proof of the migration: file count, byte count, and operation duration — a compliant paper trail

Migrations that omit /SEC or /COPYALL silently drop NTFS ACLs. A file restricted to DOMAIN\Finance on the source can arrive at the destination with default "Everyone: Read" permissions—a direct cloud misconfiguration.

Best practice: always run a /L dry-run first to audit what will be transferred, then validate destination permissions after migration. Confirm that the destination volume is a native NTFS volume before using /SEC.

6. Scenario 3 — Forensic Log Collection (Digital Forensics)

Context:

A SOC analyst receives an alert indicating a possible brute-force attack against a privileged account. The analyst must collect log files for SIEM ingestion and forensic analysis under two constraints: (1) only log files should be collected (scope limitation), and (2) original file timestamps must be preserved (chain-of-custody integrity).

Collect only .log files — preserving full metadata:

```
robocopy C:\CyberLab\Staging C:\CyberLab\ForensicsZone *.log /S /COPYALL
/LOG:C:\CyberLab\chain_of_custody.log
```


FORENSIC COLLECTION COMMAND: *.LOG FILTER WITH /COPYALL AND AUDIT LOG

```
C:\>robocopy C:\CyberLab\Staging C:\CyberLab\ForensicsZone *.log /S /COPYALL
/LOG:C:\CyberLab\chain_of_custody.log

Log File : C:\CyberLab\chain_of_custody.log
```

Flag reference: *.log — collect only files with the .log extension across all matching directories; /S — copy subdirectories but not empty ones (/E would include empty directories); /COPYALL is equivalent to /COPY:DATSOU and copies: Data, Attributes, Timestamps, Security ACLs, Owner information, and Auditing information.

Verify collected evidence — directory listing with timestamps:

DIR FORENSICSZONE /S: LOGS\ DIRECTORY WITH AUTH.LOG AND SYSTEM.LOG — ORIGINAL TIMESTAMPS PRESERVED

```
C:\>dir C:\CyberLab\ForensicsZone /S
Volume in drive C is Windows
Volume Serial Number is C48C-B006

Directory of C:\CyberLab\ForensicsZone
06/05/2026  01:24 AM    <DIR>          .
06/05/2026  01:24 AM    <DIR>          ..
06/04/2026  10:33 PM    <DIR>          Logs
                0 File(s)                0 bytes

Directory of C:\CyberLab\ForensicsZone\Logs
06/04/2026  10:33 PM    <DIR>          .
06/05/2026  01:24 AM    <DIR>          ..
06/04/2026  10:33 PM             430 auth.log
06/04/2026  10:37 PM             57 system.log
                2 File(s)                487 bytes

Total Files Listed:
                2 File(s)                487 bytes
                5 Dir(s) 55,296,946,176 bytes free
```

The directory listing confirms that only the Logs\ subdirectory was created within ForensicsZone\, and only auth.log and system.log were collected. No configuration files, user documents, or temporary files were copied, satisfying the scope constraint. Crucially, the file timestamps reflect the original creation dates—not the time of collection.

Chain-of-custody record:

CHAIN_OF_CUSTODY.LOG: COLLECTION TIMESTAMP, SOURCE, DESTINATION, AND FILE MANIFEST

```
C:\>type C:\CyberLab\chain_of_custody.log

-----
ROBOCOPY      ::      Robust File Copy for Windows
-----

Started : Friday, June 5, 2026 1:24:14 AM
Source  : C:\CyberLab\Staging\
Dest    : C:\CyberLab\ForensicsZone\

Files : *.log

Options : /S /COPYALL /R:1000000 /W:30

-----

New Dir.      0      C:\CyberLab\Staging\
New Dir      0      C:\CyberLab\Staging\Configs\
New Dir      2      C:\CyberLab\Staging\Logs\
100% New File      430      auth.log
100% New File      57      system.log
New Dir      0      C:\CyberLab\Staging\Temp\
New Dir      0      C:\CyberLab\Staging\UserData\
```

Timestamp comparison: auth.log was created on 06/04/2026 at 10:33 PM. The robocopy collection was performed on Friday, 06/05/2026 at 1:24:14 AM. The original creation timestamp is intact in the collected evidence, despite the file being copied over 3 hours later.

Observations:

- Only .log files were collected — no configuration files, user documents, or temporary files were included
- auth.log reveals a clear brute-force pattern: five AUTH_FAILED entries from 185.220.101.5 within five seconds, followed by ACCOUNT_LOCKED — consistent with automated credential-stuffing tooling
- The IP range 185.220.101.x corresponds to known Tor exit node infrastructure
- chain_of_custody.log documents the exact time of collection, source and destination paths, and the file manifest — admissible as a collection record

Drag-and-drop copying in Windows Explorer silently updates the Date Modified timestamp, destroying forensic integrity. /COPYALL preserves original timestamps, owner information, and ACLs—making robocopy a valid first-response evidence collection tool when dedicated forensic platforms (FTK, Autopsy, Velociraptor) are not immediately available.

The combination of /COPYALL and a /LOG output file is functionally equivalent to a documented forensic acquisition: the log serves as a collection manifest, and the preserved metadata supports chain-of-custody requirements.

7. Conclusion

This lab demonstrated three practical applications of the Windows robocopy utility, approached from the distinct perspectives of penetration testing, cloud security, and digital forensics.

From the penetration testing standpoint, the /L flag proved that a threat actor with read access to a file share can silently enumerate its entire contents—identifying high-value targets such

as API keys, database credentials, and live authentication tokens—without copying a single byte or triggering write-based access alerts. Understanding this enumeration technique is equally valuable to defenders, who should audit read permissions on sensitive shares with the same rigour applied to write permissions.

From the cloud security standpoint, the migration exercise highlighted a frequently overlooked risk: NTFS ACL loss during infrastructure moves. The deliberate error using the VM shared folder (Z:\CloudMigration) demonstrated that when a destination volume does not support persistent ACLs, a migration can fail or even appear to succeed while silently stripping access controls from every transferred file. Migrating to a native NTFS volume with the /SEC flag resolved this, confirming that destination selection and permission validation are as critical as the migration command itself.

From the digital forensics standpoint, the /COPYALL flag preserved the original creation timestamp of auth.log (June 4, 2026) through a collection run performed on June 5, 2026—demonstrating the timestamp integrity required for a defensible chain of custody. The auth.log content simultaneously illustrated a recognisable brute-force pattern: five consecutive authentication failures from the same IP address within five seconds, followed by an account lockout. This combination of timestamp preservation and log analysis reflects the first-response evidence collection workflow employed when dedicated forensic platforms (FTK or Autopsy) are unavailable.

Robocopy ships with every Windows installation. In constrained environments where PowerShell is unavailable or policy-restricted—such as secure boot configurations or hardened enterprise workstations—it remains a reliable, auditable, and powerful tool for enumeration, compliant data migration, and forensic evidence collection alike.